

MUSIE

A-to-Z Repair & Maintenance Guide

YouTube Music, Spotify, Listen Together, local build, phone testing, GitHub push, release, updater and disaster recovery

Made for a non-coder

Every important action is shown in simple Hinglish with copy-paste PowerShell commands. Follow the STOP boxes strictly.

Project: Musie 0.8.12 (versionCode 28)

Repository: github.com/dipkp/Musie

Prepared for: Dipraksh / dipkp

Platform: Windows + Android + GitHub Actions

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1. Sabse pehle: is guide ka correct use

Ye guide tab use karni hai jab Musie mein YouTube playback, Spotify login/API, Listen Together backend, download, like sync, lyrics ya GitHub release/updater toot jaye. Iska goal Musie ka custom interface aur branding bachakar backend/engine fix port karna hai.

Reality check

Koi bhi guide external YouTube, Spotify ya third-party server ke future changes ke against 100% lifetime guarantee nahi de sakti. Lekin exact diagnosis, minimal patch aur complete testing follow karne par galat copy-paste ka risk bahut kam ho jata hai.

Golden rules

- Meld ka poora repository Musie ke upar kabhi paste mat karo.
- Musie ki UI, logo, fonts, player, navigation, About page aur preferences ko preserve karo.
- Har repair ke liye alag fix/... branch banao.
- Pehle logs se root cause pakdo; uske baad exact Meld commit ya exact files dekho.
- Build successful hona functional success nahi hota. Phone par real play/download/like test compulsory hai.
- Keystore, passwords, cookies, Spotify tokens aur GitHub PAT kabhi commit ya screenshot mein share mat karo.

Musie ki current verified identity

Item	Current value
GitHub	https://github.com/dipkp/Musie
App	Musie
Package ID	com.meld.app (release); debug suffix normally .debug
Version	0.8.12, versionCode 28
JDK	21
Variants	FOSS, GMS (Google Cast), Izzy
Primary local build	.\gradlew.bat :app:assembleFossDebug
Latest public release	v0.8.12

2. Architecture samjho: front-end aur backend alag layers

Musie ka custom front-end Meld se alag ho sakta hai, lekin backend fixes phir bhi port ho sakte hain. Condition ye hai ki interface contract - function parameters, return models, player state aur database calls - match kare.

Layer	Examples	Repair rule
UI/front-end	Compose screens, player, navigation, artwork, theme	Normally Musie version keep karo.
ViewModel/state	Screen state, loading, error, user actions	API signature badle to adapter update karo.
Service/player	MusicService, PlayerConnection, queues	Playback sync fixes carefully integrate karo.
Provider engine	Innertube/YouTube, Spotify client	Meld ke relevant engine commits port ho sakte hain.
Remote backend	Listen Together WebSocket, hash registry	Server dead ho to APK-only fix sufficient nahi hoga.
CI/release	GitHub Actions, signing, APK assets	App logic se separate diagnose karo.

UI dikh rahi hai par button kaam nahi karta

Iska matlab aksar UI layer sahi hai, lekin onClick -> ViewModel -> provider/service chain mein koi link fail hai. Har link ko logs aur breakpoint/temporary logging se verify karna hota hai.

Main dependency flow

```

Tap Play / Like / Download
|
v
Compose UI callback
|
v
ViewModel or PlayerConnection
|
v
MusicService / Database / Provider client
|
v
YouTube / Spotify / Listen Together server

```

3. Disaster recovery: laptop ki sari files delete ho jaye to

3.1 Jo cheezein alag backup karni hi hongi

- GitHub account access aur 2FA recovery codes.
- Android signing keystore file, alias aur passwords - encrypted USB/cloud vault mein.
- GitHub Actions secrets ki list. Existing repo mein secrets survive karte hain, lekin GitHub unki original values download nahi karne deta.
- Optional API keys: LASTFM_API_KEY, LASTFM_SECRET, SPOTIFY_CLIENT_ID, crash reporting token.
- Phone par installed working APK ka offline copy.

Critical

Signing keystore lost hua to same package ki existing installation ko future APK se update karna impossible ho sakta hai. User ko uninstall karke fresh install karna padega, jisse local data lose ho sakta hai.

3.2 Fresh computer par software

- Git for Windows
- Android Studio with Android SDK 36 and Build Tools 35
- JDK 21 (Android Studio embedded JDK ya Temurin 21)
- Platform Tools (adb)
- PowerShell 7 optional; Windows PowerShell bhi chalega

3.3 Clone Musie

```
cd %env:USERPROFILE%\Documents
git clone https://github.com/dipkp/Musie.git Musie
cd Musie
git submodule update --init metroproto
git status
git remote -v
```

Expected: branch main, origin dipkp/Musie, aur working tree clean. Private notes submodule accessible na ho to usko force mat karo; build ke liye metroproto important hai.

3.4 Original Meld ko upstream naam se add karo

```
git remote add upstream https://github.com/FrancescoGrazioso/Meld.git
git fetch origin --prune
git fetch upstream --prune
git remote -v
```

Meaning

origin = aapka Musie repo. upstream = original Meld. Push hamesha origin par. Upstream se sirf fixes inspect/fetch karne hain.

4. Har repair se pehle safe branch aur snapshot

1. Main ko latest karo.

```
git switch main
git pull --ff-only origin main
git status
```

2. Repair branch banao. Example:

```
git switch -c fix/youtube-playback
# ya: git switch -c fix/spotify-login
# ya: git switch -c fix/listen-together-sync
```

3. Current recovery point note karo.

```
git rev-parse HEAD
git log -5 --oneline
git diff --stat
```

STOP

Agar git status mein unknown modified files dikh rahe hain, pehle samjho ki ye kisne badle. Kisi aur ke changes reset/delete mat karo.

5. Problem ko classify karo - andaza mat lagao

Symptom	Likely area	First proof
Search result aata hai, play 403/2000	YouTube player/cipher/client	Logcat + stream URL response
Search bhi blank	Browse/API client/response model	HTTP status and parse exception
Spotify login insecure browser	Auth/WebView/provider policy	Login screen URL + WebView logs
Spotify playlist open but actions fail	GQL hash/token/mutation	HTTP 401/412 and operation name
Listen Together connected, guest silent	Player bridge/state timing/server event	Both device logs with room event
App opens then stops	Runtime crash/ANR	FATAL EXCEPTION or ANR traces
APK builds but update fails	Signing/package/version/release JSON	Package ID, cert, tag_name, versionCode

Evidence collection on phone

```
adb devices
adb logcat -c
# Ab phone par problem reproduce karo.
adb logcat -d > musie-log.txt

# Useful filters:
adb logcat -d | Select-String -Pattern "FATAL
EXCEPTION|ANR|403|401|412|2000|IO_UNSPECIFIED|Spotify|YouTube|ListenTogether"
```

Privacy

Log share karne se pehle Authorization headers, cookies, sp_dc, sp_key, access_token, refresh_token, passwords, email aur private room codes redact karo.

6. Meld mein fix dhoondhne ka safe method

1. Upstream update fetch karo, Musie code change kiye bina.

```
git fetch upstream --prune
```

2. Recent commits keyword se search karo.

```
git log upstream/main --oneline --all --grep="youtube"  
git log upstream/main --oneline --all --grep="spotify"  
git log upstream/main --oneline --all --grep="cipher"  
git log upstream/main --oneline --all --grep="playback"  
git log upstream/main --oneline --all --grep="listen together"
```

3. Candidate commit ko poora inspect karo.

```
git show --stat <COMMIT_HASH>  
git show <COMMIT_HASH>  
git diff <COMMIT_HASH>^ <COMMIT_HASH> -- path/to/relevant/file.kt
```

4. Check karo candidate commit kin Musie-custom files ko touch karta hai.

```
git diff --name-only main...upstream/main  
git diff --name-only <COMMIT_HASH>^ <COMMIT_HASH>
```

Decision

Agar fix sirf backend module mein hai aur Musie ne file customize nahi ki, exact commit cherry-pick possible hai. Agar custom file involved hai to manual minimal port karo; entire file replace mat karo.

7. YouTube Music repair playbook

7.1 Exact important files

- innertube/src/main/kotlin/com/metrolist/innertube/models/YouTubeClient.kt - client identities/capabilities
- innertube/src/main/kotlin/com/metrolist/innertube/YouTube.kt - Innertube requests and responses
- app/src/main/kotlin/com/metrolist/music/utis/YTPlayerUtils.kt - playable formats and player selection
- app/src/main/kotlin/com/metrolist/music/utis/cipher/ - signature and n-transform extraction
- innertube/src/test/kotlin/com/metrolist/innertube/PlaybackForbiddenDiagnosticsTest.kt - playback diagnostics

7.2 Symptom to root-cause map

Error	Meaning	Typical fix
IO_UNSPECIFIED (2000)	Player could not open returned stream	Fresh player response, valid URL, client/cipher fix
HTTP 403	URL/signature/client rejected	Cipher/n-transform or client identity update
No playable format	Formats empty/restricted	Client fallback and response handling
Bot/login required	YouTube risk/auth response	Cookie/auth handling or alternate supported client
Only preview plays	Returned URL is limited/expired	Validate duration and request fresh stream
Search works, playback fails	Browse API OK; player pipeline broken	Do not replace browse UI; fix player path

7.3 Repair sequence

1. Reproduce one known public song on stable Wi-Fi and mobile data.
2. Capture logcat and note exact HTTP/status/player response.
3. Run existing playback diagnostic test if available.
4. Fetch Meld upstream and locate a commit addressing the same error, not merely a similar UI symptom.
5. Port provider/client/cipher changes first. Preserve Musie UI and preferences.
6. If models changed, update only affected call sites until compile succeeds.
7. Test search -> select -> first play -> seek -> next -> replay -> like -> download.
8. Test at least three tracks: normal music video, audio/official track, age/region-sensitive content if legally available.

7.4 Build/test commands

```
.\gradlew.bat :innertube:test --stacktrace
.\gradlew.bat :app:assembleFossDebug --stacktrace

Get-ChildItem app\build\outputs\apk -Recurse -Filter *.apk
adb install -r "FULL_PATH_TO_APK"
```

Do not accept false success

Player API status OK ya search result dikhna enough nahi. Actual media URL phone par open ho, seek chale aur preview limit ke baad bhi playback continue ho - tab pass.

8. Spotify repair playbook

8.1 Exact important files

- spotify/src/main/kotlin/com/metrolist/spotify/SpotifyAuth.kt
- spotify/src/main/kotlin/com/metrolist/spotify/Spotify.kt

- spotify/src/main/kotlin/com/metrolist/spotify/SpotifyHashProvider.kt
- app/src/main/kotlin/com/metrolist/music/utils/SpotifyTokenManager.kt
- app/src/main/kotlin/com/metrolist/music/utils/SpotifyHashSync.kt
- app/src/main/kotlin/com/metrolist/music/ui/screens/SpotifyLoginScreen.kt
- docs/spotify-gql-hashes.json
- .github/workflows/spotify-hash-check.yml

8.2 Four failures ko alag samjho

Failure	Evidence	Fix area
Login blocked/insecure browser	Provider page refuses WebView/social sign-in	Auth flow; direct email login guidance; trusted browser only if supported
Token expired	401/unauthorized	Refresh/session handling
GQL hash rotated	412 or operation failure	Hash registry and fallback hash
Response schema changed	Parsing/serialization exception	Data models and mapping

Current external dependency

Musie currently reads Spotify hashes from <https://francescograzioso.github.io/Meld/spotify-gql-hashes.json>. Long-term independence ke liye same JSON ko dipkp/Musie GitHub Pages par serve karke SpotifyHashSync URL apne endpoint par badalna better hai.

8.3 Spotify repair sequence

1. Decide: login fail, token fail, hash fail, ya parsing fail.
2. Network/logcat mein operation name aur HTTP status note karo; tokens redact karo.
3. Meld upstream ke exact corresponding commit ko inspect karo.
4. Auth fix ko hash fix ke saath unnecessarily mix mat karo.
5. New GQL operation add ho to Spotify.kt aur docs/spotify-gql-hashes.json dono update karo.
6. Test login, profile, search, playlist open, liked songs, add/remove like, add/remove playlist item.
7. Logout/login repeat test karo; app restart ke baad token persistence test karo.

9. Listen Together repair playbook

Listen Together teen systems ka combination hai: room server, protocol/client, aur Android player bridge. Dono devices par Connected dikhne ke baad bhi guest playback silent ho sakta hai agar PLAY event player ready hone se pehle aa jaye, media item prepare na ho, ya server event forward na kare.

9.1 Important files

- app/src/main/kotlin/com/metrolist/music/listentogether/ListenTogetherClient.kt
- app/src/main/kotlin/com/metrolist/music/listentogether/ListenTogetherManager.kt
- app/src/main/kotlin/com/metrolist/music/listentogether/Protocol.kt
- app/src/main/kotlin/com/metrolist/music/listentogether/MessageCodec.kt
- app/src/main/kotlin/com/metrolist/music/listentogether/ListenTogetherServers.kt
- app/src/main/kotlin/com/metrolist/music/service/MusicService.kt
- app/src/main/kotlin/com/metrolist/music/playback/PlayerConnection.kt
- app/src/main/kotlin/com/metrolist/music/MainActivity.kt

Current default external WebSocket server: `wss://metroserverx.meowery.eu/ws`. Server aapke APK ke bahar hai.

9.2 Two-device test is mandatory

Ek physical phone ho to second device Android emulator ho sakta hai. Laptop se phone connect karna logs/install ke liye useful hai, lekin laptop automatically second Listen Together client nahi banta.

1. Device A host room create kare; Device B same room code join kare.
2. Dono mein Connected and same member count verify karo.
3. Host fresh song play kare. Guest exact same media ID receive kare.
4. Guest STATE_READY tak wait kare, media items set/prepare hon, phir seek/play apply ho.
5. Pause, resume, seek, next, previous aur late join test karo.
6. Host background/foreground aur screen off/on test karo.
7. Dono devices ka separate timestamped log save karo.

Server dead case

WebSocket server unavailable ho to APK code copy karke fix nahi hoga. Apna compatible server deploy karke ListenTogetherServers.kt/default settings ko new `wss://` URL par point karna padega.

10. Minimal port: exact code kaise Musie mein lao

Method A - clean backend-only commit

```
git show --stat <COMMIT_HASH>
git cherry-pick -n <COMMIT_HASH>
git diff --stat
git diff
```

-n commit create nahi karta. Pehle diff review karo. UI/branding files touch ho rahe hon to restore nahi, balki manual edit se unwanted hunks hatao. Changes correct hon tab human approval ke baad commit.

Method B - custom file involved (recommended)

1. Meld commit ka diff side-by-side kholo.
2. Changed function/class ka purpose note karo.
3. Musie file mein corresponding function locate karo.

4. Sirf required imports, data fields, branch logic aur call-site changes apply karo.
5. Musie composables, colors, navigation, strings, branding aur preferences untouched rakho.
6. Compile after small group of edits; ek saath dozens files mat badlo.

Useful comparison commands

```
git show upstream/main:path/to/file.kt > $env:TEMP\meld-file.kt
git diff --no-index path\to\musie-file.kt $env:TEMP\meld-file.kt
git diff --word-diff main...upstream/main -- path/to/file.kt
rg -n "FunctionName|ClassName|error text" app innertube spotify
```

Never do this

Meld folder ko Musie ke upar copy/paste, git checkout --theirs on whole files, git reset --hard, ya hundreds of commits ka blind merge mat karo.

11. Local build from zero

11.1 Environment verify

```
java -version
adb version
git --version
.\gradlew.bat --version
```

Java output 21 hona chahiye. Agar wrong JDK ho, Android Studio Settings -> Build Tools -> Gradle -> Gradle JDK mein JDK 21 select karo, ya JAVA_HOME ko JDK 21 par set karo.

11.2 Fast development build

```
.\gradlew.bat :app:assembleFossDebug --stacktrace
```

Expected APK: app/build/outputs/apk/universalFoss/debug/app-universal-foss-debug.apk. Exact generated path find karne ke liye:

```
Get-ChildItem app\build\outputs\apk -Recurse -Filter *.apk | Select-Object
FullName,Length,LastWriteTime
```

11.3 Build error reading order

- Log ke last 80-150 lines padho, first useful 'What went wrong' identify karo.
- Warning ko failure mat samjho. Example unnecessary safe call normally non-fatal warning hai.
- Keytool/keystore error = signing problem, source-code problem nahi.
- Unresolved reference/type mismatch = code port incomplete.
- KSP/Room schema error = model/database compile issue.
- R8 release-only error = keep rules or release optimization issue.

11.4 Common keystore error

Error Tag number over 30 is not supported usually invalid/corrupted keystore bytes indicate karta hai. Base64 secret ko correct binary keystore mein decode karo; text file ko .keystore rename karna kaam nahi karega.

```
keytool -list -v -keystore path\to\release.keystore
# Password prompt par correct store password enter karo.
```

12. Phone installation and proper testing

12.1 Device connect

```
adb devices
# Phone prompt: Allow USB debugging
adb install -r "FULL_PATH_TO_APK"
```

Agar INSTALL_FAILED_UPDATE_INCOMPATIBLE aaye, installed APK ka signature different hai. Data important ho to uninstall mat karo. Same signing keystore se APK sign karo.

12.2 Clean test vs update test

Test	Purpose	Action
Update install	Existing user data and migration	adb install -r; do not uninstall
Clean install	First-run defaults/welcome	Backup data, uninstall only with permission, install fresh
Offline restart	Cache/local library	Airplane mode then reopen
Process death	State restoration	Force stop, reopen

12.3 Regression checklist - release se pehle sab PASS

- [] Cold start and warm start without 'Musie keeps stopping'
- [] Home slider manual swipe; no duplicate same recommendation
- [] Search, select and play
- [] Play/pause/seek/next/previous
- [] Queue, sleep timer, shuffle, lyrics and repeat buttons
- [] Double-tap artwork like; unlike; liked list refresh
- [] Download, downloaded playback, delete download
- [] Playlist create, custom artwork, add/remove/reorder
- [] YouTube account playlists and sync
- [] Spotify login, search, library, add/remove like
- [] Lyrics first load, cached load, auto-scroll and tap positioning
- [] Mini-player and full-player all background styles
- [] Navigation home/search/playlist and back gesture
- [] Listen Together two-device host/guest full controls
- [] Updater detects a genuinely higher public GitHub release

13. Git commit, push and review

Ye commands tab hi run karo jab diff reviewed, local build pass aur phone tests pass ho. AGENTS.md ke hisab se AI ko commit/push/merge/version bump ke liye explicit human instruction chahiye.

```
git status
git diff --check
git diff --stat
git diff

git add path/to/changed/file1 path/to/changed/file2
git commit -m "fix(youtube): restore stream playback"
git push -u origin fix/youtube-playback
```

Before push

git status mein .android-sdk, .gradle caches, local.properties, keystore, logs, tokens ya generated APK staged nahi hone chahiye.

Merge after review

```
git switch main
git pull --ff-only origin main
git merge --no-ff fix/youtube-playback
git push origin main
```

Agar GitHub Pull Request use kar rahe ho to local merge ki zarurat nahi; PR checks pass hone ke baad GitHub UI se merge kar sakte ho.

14. Version bump and GitHub release

Current verified version 0.8.12/code 28 hai. Next normal release example 0.8.13/code 29 hoga, lekin pehle repository ka current value dobara check karo. Version bump sirf explicit owner approval par.

```
Select-String -Path app\build.gradle.kts -Pattern "versionCode|versionName"
```

Approved ho to app/build.gradle.kts mein dono values increment karo. Repository workflow app/build.gradle.kts change on main ya manual workflow_dispatch par FOSS, GMS aur Izzy release builds banata hai, sign karta hai aur GitHub Release create/update karta hai.

Three APKs ka difference

Asset	Use
Musie.apk	FOSS/default; built-in updater enabled; no Google Cast
Musie-with-Google-Cast.apk	GMS; Google Cast supported; updater enabled
Musie-izzy.apk	IzzyOnDroid-style; no Cast; built-in updater disabled

Manual workflow run

GitHub -> Musie -> Actions -> Bump to new version -> Run workflow -> main. Build ke teen matrix jobs aur final create-release job green hone chahiye.

CLI alternative, authenticated gh ho to:

```
gh workflow run release.yml --repo dipkp/Musie --ref main
gh run list --repo dipkp/Musie --workflow release.yml --limit 5
gh run watch <RUN_ID> --repo dipkp/Musie
```

Required GitHub Actions secrets

- KEYSTORE - base64 encoded signing keystore
- KEY_ALIAS
- KEYSTORE_PASSWORD
- KEY_PASSWORD
- Optional: LASTFM_API_KEY, LASTFM_SECRET, CRASH_REPORT_TOKEN

Release visibility

Release public hona chahiye, draft/prerelease nahi, aur tag vX.Y.Z format mein. Varna /releases/latest expected tag_name nahi dega aur in-app updater fail ho sakta hai.

15. In-app updater end-to-end verification

Musie updater GitHub endpoint <https://api.github.com/repos/dipkp/Musie/releases/latest> se latest release padhta hai. JSON mein tag_name required hai.

```
$release = Invoke-RestMethod https://api.github.com/repos/dipkp/Musie/releases/latest
$release.tag_name
$release.name
$release.draft
$release.prerelease
$release.assets | Select-Object name,browser_download_url,size
```

Pass conditions:

- tag_name expected higher version ho, e.g. v0.8.13
- draft = false and prerelease = false
- Correct APK asset exists
- Old installed app ka versionCode latest se lower ho
- FOSS/GMS build ho; Izzy variant mein updater intentionally disabled hai
- Package ID and signing certificate same ho

16. Rollback plan - bad release ko safely reverse karo

1. Bad release ka exact commit aur tag note karo.

```
git log -10 --oneline
git tag --list --sort=-version:refname | Select-Object -First 10
```

2. History rewrite nahi. Bad change ka revert commit banao.

```
git switch -c fix/revert-bad-release main
git revert <BAD_COMMIT_HASH>
```

3. Build and full regression test. Then new higher version release karo. Existing published tag ko silently replace karna avoid karo.

User safety

Android lower versionCode ko update ke roop mein install nahi karega. Rollback code ko new higher versionCode ke saath release karna hota hai.

17. Simple PowerShell automation helpers

In commands ko manually run kar sakte ho. Inhe blindly permanent script banane se pehle paths verify karo.

One-command health snapshot

```
git status
git remote -v
git branch --show-current
java -version
adb devices
Select-String -Path app\build.gradle.kts -Pattern "versionCode|versionName"
```

Build, locate and install latest FOSS debug APK

```
.\gradlew.bat :app:assembleFossDebug --stacktrace
$apk = Get-ChildItem app\build\outputs\apk -Recurse -Filter *.apk |
Sort-Object LastWriteTime -Descending | Select-Object -First 1
$apk.FullName
adb install -r $apk.FullName
```

Capture focused logs

```
adb logcat -c
Write-Host "Phone par bug reproduce karo, phir Enter dabao"
Read-Host
adb logcat -d > musie-log.txt
Select-String -Path musie-log.txt -Pattern "FATAL
EXCEPTION|ANR|403|401|412|IO_UNSPECIFIED|ListenTogether"
```

18. AI ko kya prompt dena hai - copy/paste templates

AI safety instruction

AI ko pehle repository ka AGENTS.md poora read karne bolo. Explicitly bolo: no whole upstream merge, no UI overwrite, no commit/push/version/release until I approve.

18.1 Master repair prompt

You are repairing my Android app Musie in the local repository. I am not a coder.
First read AGENTS.md completely and inspect git status, current branch, remotes and version.
Origin is dipkp/Musie and upstream is FrancescoGrazioso/Meld.

Problem: [PASTE EXACT SYMPTOM]
Logs: [PASTE REDACTED RELEVANT LOGS]

Requirements:

1. Diagnose the root cause before editing.
2. Fetch/inspect Meld upstream and identify the exact relevant commit(s).
3. Do not merge or copy the whole Meld repository.
4. Preserve all Musie UI, branding, player, navigation, fonts and settings.
5. Port only the minimum backend/service/model changes needed.
6. Create a fix/... branch. Do not discard existing user changes.
7. Show me changed files and explain why each is needed.
8. Ask before running a full build if AGENTS.md requires it.
9. After approval, build :app:assembleFossDebug and report the exact APK path.
10. Give me a phone regression checklist for this bug.
11. Do not commit, push, merge, bump version or release until I explicitly ask.
12. Never expose tokens, cookies, signing passwords or keystores.

18.2 YouTube playback prompt

Musie search/browse works, but playback fails with [403 / IO_UNSPECIFIED 2000 / no playable format]. Trace the call from UI to PlayerConnection/MusicService, YTPlayerUtils, Innertube player response and cipher/n-transform. Compare the exact Meld upstream fix. Prove the returned stream URL actually plays beyond any preview window. Preserve Musie's UI. Test play, seek, next, like and download.

18.3 Spotify prompt

Classify this Spotify failure as login policy, token refresh, GQL hash rotation or response-schema change. Inspect SpotifyAuth.kt, Spotify.kt, SpotifyHashProvider.kt, SpotifyTokenManager.kt, SpotifyHashSync.kt and SpotifyLoginScreen.kt. Record HTTP status and operation name without exposing tokens. Port only the matching Meld fix. Test profile, search, liked songs, playlist mutations and logout/login persistence.

18.4 Listen Together prompt

Both clients show Connected, but guest playback does not start/sync. Determine whether the server forwards events and whether the guest sets media items, prepares, waits for STATE_READY, then seeks/plays. Inspect ListenTogetherClient/Manager, protocol/codecs, PlayerConnection, MusicService and MainActivity lifecycle. Do not change the Musie UI. Require a two-client test including late join and backgrounding.

18.5 Build error prompt

Read the complete build failure, separate warnings from the first real error, and identify whether it is source code, dependency, SDK/JDK, signing, R8 or CI. Do not make unrelated edits. Fix the smallest root cause and rerun only the failed/closest task first. Then ask before a full app build. Here is the log: [PASTE LOG]

19. Success probability and release gate

Agar Meld ne exactly same problem already fix ki hai aur Musie architecture nearby version par hai, careful minimal port ka practical success chance high hota hai. Lekin percentage proof nahi hota; real proof tests hain.

Stage	What it proves	What it does not prove
Code compiles	Types/imports/build graph valid	Playback, login, server and buttons work
APK installs	Package/signature accepted	Runtime feature behavior
One song plays	Basic player path	All clients, download, like, next track
Regression suite passes	High confidence for tested device/network	Future external API stability
Public updater test passes	Release JSON + asset + version path	Every user's OEM behavior

Rough expectation when an exact upstream fix exists: careful port 90-97%; before full tests

UI-working-but-action-failing risk can still be around 5-15% depending on feature. After clean build plus complete device regression, serious remaining release risk can often be reduced to roughly 1-5%, but never absolute zero.

Final release gate - all must be YES

- [] Root cause proven with logs/test
- [] Exact diff reviewed; no whole-file UI overwrite
- [] Secrets and keystore not committed
- [] FOSS debug build successful
- [] Update-install test successful
- [] Clean-install/first-run test successful when relevant
- [] All affected actions function, not just display
- [] No crash/ANR during 15+ minute mixed use
- [] Two-device Listen Together test if related
- [] GitHub Actions release jobs green
- [] Latest-release API returns correct public tag and APK assets

20. One-command repair-check automation

The repository can include tools/Musie-Repair-Check.ps1. This PowerShell helper saves one timestamped report containing repository status, tools, current version, relevant Meld comparison, latest APK and optional public-release check. It is designed to make diagnosis fast before an AI or developer changes any code.

What it does

It can check the current repo, detect common error categories from a redacted log, optionally fetch Meld upstream, list relevant upstream commits/files, build a FOSS debug APK if requested, find the APK, install it if requested, and verify the public GitHub latest-release endpoint.

What it never does

It intentionally does not copy Meld code, automatically apply a commit, delete data, change versionCode, push to GitHub, create a release or expose secrets. Those actions require review because a generic blind repair can break Musie.

20.1 First-time setup

```
cd C:\Users\YOUR_NAME\Documents\Musie
Set-ExecutionPolicy -Scope Process Bypass
```

The Process scope affects only the current PowerShell window. It does not permanently change Windows policy.

20.2 Common one-command runs

```
# General health + updater/release check
.\tools\Musie-Repair-Check.ps1 -Scope Auto -CheckLatestRelease

# YouTube failure: compare upstream and build a test APK
.\tools\Musie-Repair-Check.ps1 -Scope YouTube -FetchUpstream -Build

# Spotify failure with a saved redacted log
.\tools\Musie-Repair-Check.ps1 -Scope Spotify -LogFile "C:\Temp\musie-log.txt" -FetchUpstream

# Listen Together connected but guest is silent
.\tools\Musie-Repair-Check.ps1 -Scope ListenTogether -LogFile "C:\Temp\host-or-guest-log.txt"
-FetchUpstream -Build

# Install the latest local APK after you already tested the build
.\tools\Musie-Repair-Check.ps1 -Scope Auto -InstallLatestApk
```

20.3 Result and next action

Every run creates repair-reports\YYYYMMDD-HHMMSS\Musie-Repair-Check.txt. Give that file - after redacting tokens/cookies - together with the relevant error log to an AI. The report tells it where to inspect; it is not a guarantee that an upstream change is compatible.

```
# Ask an AI after the report finishes:
Read repair-reports\YYYYMMDD-HHMMSS\Musie-Repair-Check.txt and my redacted log.
Find the exact matching Meld upstream commit. Show the minimal Musie diff needed,
preserve every Musie UI/branding change, and do not commit/push/build a release
until I approve.
```

21. One-page emergency checklist

1 - STOP

Do not paste all Meld files. Do not reset hard. Do not uninstall the working app if its data matters.

2 - BACKUP

Record current commit, branch, working APK and signing key backup.

3 - PROVE

Reproduce once, save redacted logcat, classify YouTube/Spotify/LT/build/updater.

4 - COMPARE

Fetch upstream, find exact Meld commit, inspect file/function diff.

5 - PORT

New fix branch; minimal backend changes; preserve Musie UI and branding.

6 - VERIFY

Compile, install -r, functional tests, logcat, two devices when required.

7 - SHIP

Review diff, explicit approval, push, higher versionCode, signed three-variant release, updater API check.

Owner: Dipraksh / GitHub: dipkp

Repository: <https://github.com/dipkp/Musie>

Guide scope: Musie 0.8.12 baseline. Always verify current files/version before applying commands.